MILITARY SPECIFICATION SHEET

ADAPTER, CONNECTOR, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (BETWEEN SERIES SMA TO N)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-A-55339.

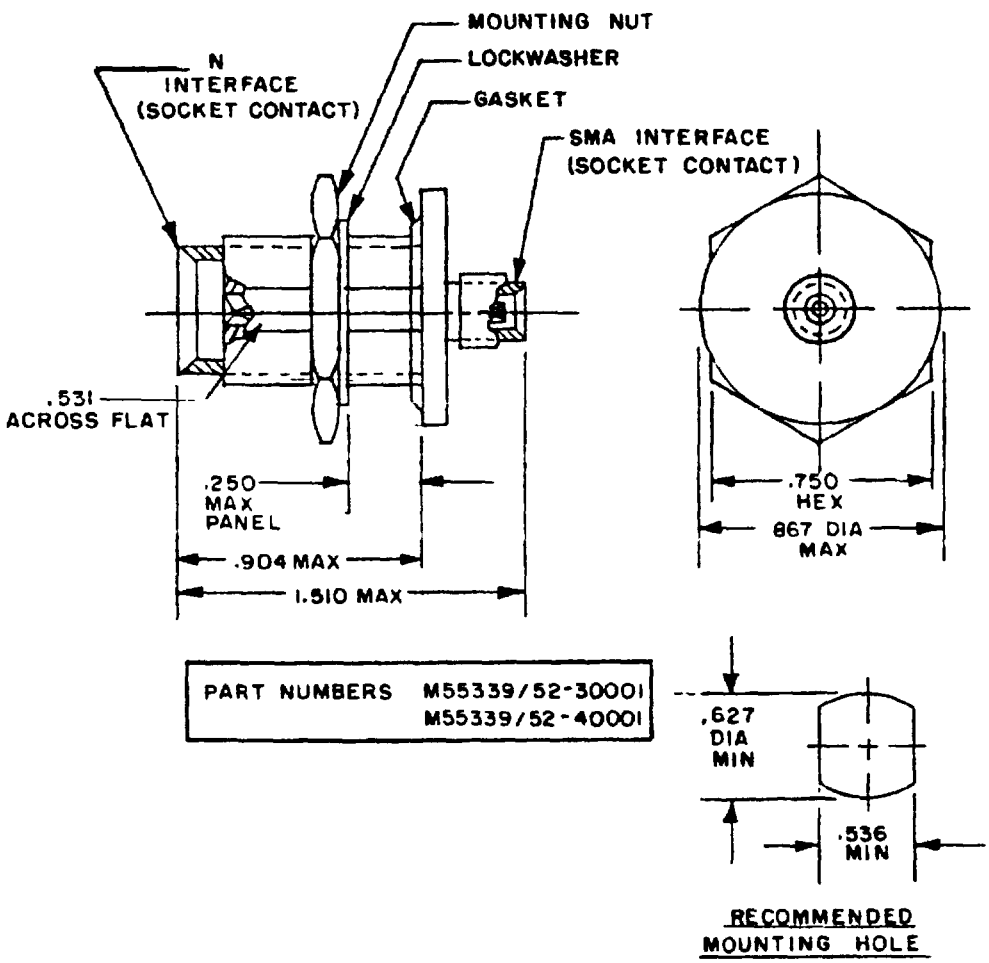
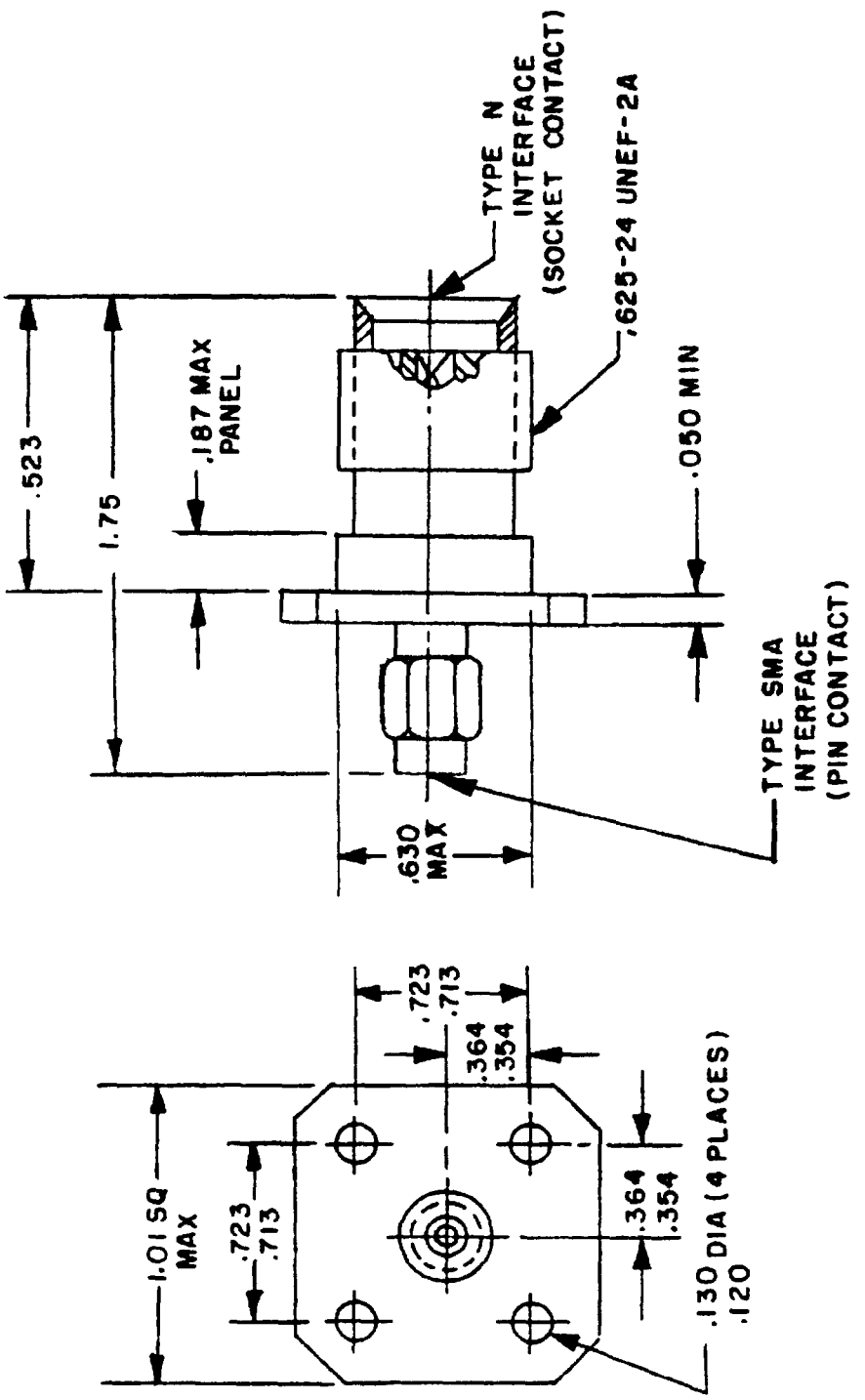


FIGURE 1. General configurations.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

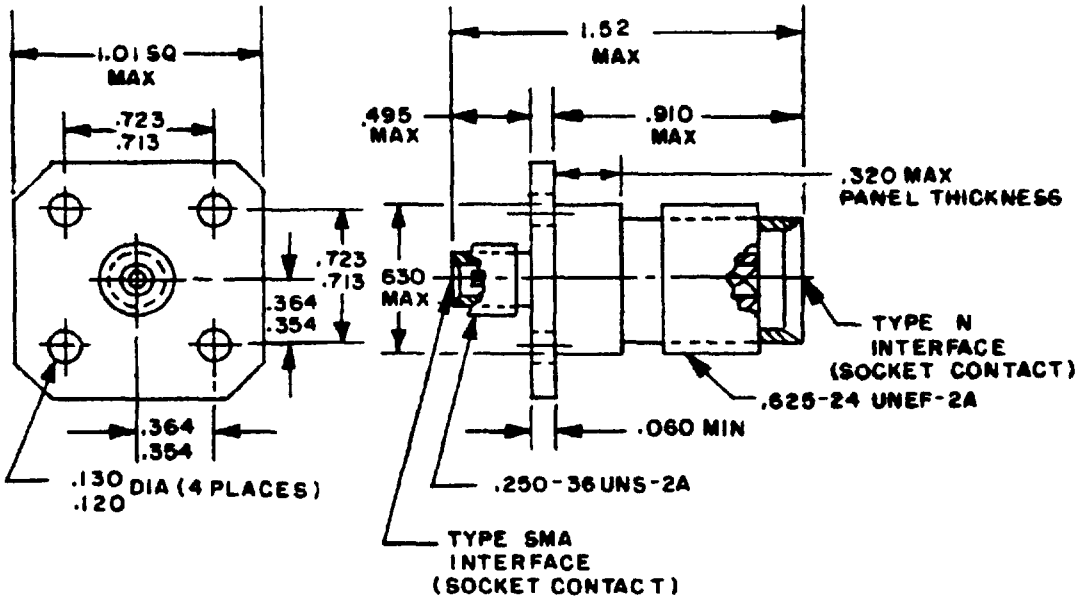
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PART NUMBERS	M55339/52-30002 M55339/52-40002

FIGURE 1. General configurations - Continued.

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PART NUMBERS	M55339/52-30003
	M55339/52-40003

Inches	mm	Inches	mm
.015	0.38	.627	15.93
.050	1.27	.630	16.00
.060	1.52	.713	18.11
.120	3.04	.723	18.36
.130	3.30	.750	19.05
187	4.75	.904	22.96
.259	6.35	.910	23.11
.320	8.13	1.01	25.7
.354	8.99	1.232	31.29
364	9.25	1.52	38.61
.495	12.57	1.75	44.45
.531	13.49		
.536	13.61		
.625	15.88		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.015 (0.38 mm).
4. All undimensioned pictorial configurations are for reference purpose only.
5. N and SMA interfaces shall be in accordance with MIL-STD-348.

FIGURE 1. General configurations - Continued.

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ENGINEERING DATA.

Nominal impedance: 50 ohms.

Frequency range: 18 GHz.

Voltage rating:

335 V rms at sea level.

85 V rms at 70,000 feet.

Operating temperature range: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Center contact retention:	Series SMA 1/	Series N
Axial force (pounds, minimum) - - - -	6	6
Torque - - - - - - - - - - - - - - - -	Not applicable	Not applicable
Coupling proof torque 2/ - - - - - - -	15 inch-pounds, minimum	
Force to engage and disengage:	Series SMA 1/	Series N
Longitudinal force (pounds, maximum) -	Not applicable	3
Torque (inch-pounds, maximum) - - - -	2	6
Mating characteristics: In accordance with MIL-STD-348.	Series SMA 1/	Series N
Center contact (socket):		
Oversize test pin diameter (inch) -	.0375 + .0001	.074 minimum
Insertion depth (inch, minimum) -	.030/.045	.125
Number of insertions - - - - - - -	3	1
Pin finish (microinches) - - - - -	16	16
Maximum test pin (insertion force test):		
Steel test pin diameter (inch, minimum)	.0370 + .0001	.066
Pin finish (microinches) - - - - - -	16	16
Insertion force (pounds, maximum) -	3	2
Insertion depth (inch, minimum) - -	.050/.075	.125

1/ Socket contact.
2/ Pin contact.

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Minimum test pin (withdrawal force):	Series SMA	Series N
Steel test pin diameter (inch)	.0355 - .0001	.063
Pin finish (microinches) - - - - - -	16	16
Withdrawal force (ounce, minimum) -	1	1
Insertion depth (inch, minimum) - -	.050/.075	125

Permeability: Less than 2.0, air = 1.0.

Seal:

Pressurized: Not applicable.

Weatherproof: Not applicable.

Insulation resistance: 5,000 megohms, minimum.

VSWR:

1.06 + .005 F (GHz); dc to 12.4 GHz.

.83 + .023 F (GHz); 12.4 to 18 GHz.

RF leakage: -65 dB, minimum, 2 to 3 GHz.

RF insertion loss: .18 dB, maximum, at 9 GHz.

Dielectric withstanding voltage: 1,000 V rms at sea level, minimum.

Contact resistance (milliohms maximum):

Contact	Initial	After environmental
Center	4.1 3/	6.0
Outer	2.2	Not applicable

Vibration, high frequency: Interruptions, 1 µs maximum, method 204 of MIL-STD-202, test condition D.

Shock: Method 213 of MIL-STD-202, test condition I.

Thermal shock: Method 107 of MIL-STD-202, test condition C.

Moisture resistance: Method 106 of MIL-STD-202.

Insulation resistance: 200 megohms, minimum, within 5 minutes after removal from humidity.

3/ Two center contacts must be mated to the center conductor under test, therefore doubling "center contact" resistance.

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Corona level:

Voltage: 250 V, minimum, at 70,000 feet, minimum.

RF high potential withstanding voltage:

RF voltage. 1,000 V rms, minimum, at 5 MHz, minimum.

Salt spray: Method 101 of MIL-STD-202, test condition B.

Connector durability: 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet mating characteristics and force to engage and disengage.

Part or identifying numbers (PIN): See figure 1

Marking: As specified in MIL-A-55339.

NOTE: When a QPL source is obtained, DESC drawing 85038 will be canceled.

CONCLUDING MATERIAL

Custodians:
Army - CR
Air Force - 85
Navy - EC
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3663-01)

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